CRITICAL VALUES FOR REGRESSION-RESIDUAL BASED COINTEGRATION TESTS

Estimated Cointegrating Regression Residual:

$$z_t = y_t - \beta_0 - \beta_1\, x_{1t} - \beta_2\, x_{2t} - \beta_3\, x_{3t} - \ldots\ldots - \beta_N\, x_{Nt}$$

Number of Variables $N+1$	Sample Size	Critical Values 10%	5%	1%
2	50	3.28	3.67	4.32
	100	3.03	3.37	4.07
	200	3.02	3.37	4.00
3	50	3.73	4.11	4.84
	100	3.59	3.93	4.45
	200	3.47	3.78	4.35
4	50	4.02	4.35	4.94
	100	3.89	4.22	4.75
	200	3.89	4.18	4.70
5	50	4.42	4.76	5.41
	100	4.26	4.58	5.18
	200	4.18	4.48	5.02
6	500	4.43	4.71	5.28

Notes and Sources: Standard Dickey-Fuller and Phillips Perron unit-root tests are applied to the regression residuals using the critical values above instead of those on the previous page, focussing entirely on the coefficients of the laggeed residual. Except for the case of 6 variables, these critical values were calculated using Monte Carlo methods by Robert F. Engle and Byung Sam Yoo and obtained from their paper "Forecasting and Testing in Co-Integrated Systems," *Journal of Econometrics*, Vol. 35, 1987, page 157. The critical values for the case of 6 variables using 500 observations were calculated by Peter C. B. Phillips and S. Ouliaris, "Asymptotic Properties of Residual Based Tests for Cointegration," *Econometrica*, Vol. 58, 1990, 165-93, and were obtained from James D. Hamilton, *Time Series Analysis*, Princeton University Press, 1994, page 766, Case 2. The complete set of Phillips-Ouliaris critical values distinguish between whether or not a constant and trend are included in the cointegrating regression. These values are so similar in the three cases to the ones calculated by Engle and Yoo, based on the inclusion of a constant but not trend, that the complexities of including them here are avoided.